

APU DAS

PhD Researcher | Semiconductor Memory and Device Reliability

College of Semiconductor Research, National Tsing Hua University

Hsinchu, Taiwan | apu.das@gapp.nthu.edu.tw | +886 910 083 544

LinkedIn: [linkedin.com/in/apu-das](https://www.linkedin.com/in/apu-das) | Google Scholar

RESEARCH PROFILE

PhD researcher studying hafnium-oxide-based ferroelectric field-effect transistors (FeFETs), with a focus on memory reliability, interface defects, and cryogenic operation. Combines device fabrication and electrical characterization from 300 K to 10 K to connect gate-stack physics with memory performance. Research spans embedded memory, neuromorphic devices, and industrial memory-development collaborations. Interested in translating experimental insights into reliable, energy-efficient semiconductor memory.

EDUCATION

Ph.D. Program, College of Semiconductor Research

National Tsing Hua University (NTHU), Hsinchu, Taiwan

2026 – Present

- Converted from the M.Sc. program to the Ph.D. program in 2026; initially enrolled at NTHU in February 2025.
- Research advisor: Prof. Sourav De. Study focus: ferroelectric memory, interface engineering, device reliability, and cryogenic computing.

Bachelor of Technology, Electronics and Communication Engineering

National Institute of Technology (NIT) Agartala, India

June 2019 – May 2023

RESEARCH EXPERIENCE

Graduate Researcher – Ferroelectric Memory Devices

College of Semiconductor Research, NTHU, Taiwan

February 2025 – Present

- Characterize hafnium-oxide FeFETs from 300 K to 10 K to investigate switching behavior and reliability for embedded and cryogenic memory.
- Perform positive-up-negative-down (PUND) measurements directly on degraded transistors' gate stacks to compare polarization response with memory-window loss.
- Obtain evidence supporting interface-driven endurance degradation: polarization remains measurable after memory-window collapse near 10^4 – 10^5 cycles in the studied devices.
- Use current–voltage (I–V), capacitance–voltage (C–V), and charge-pumping measurements to investigate charge trapping, oxygen-vacancy-related mechanisms, and cycling-induced threshold-voltage shifts.
- Study FeFET applications in spike-timing-dependent plasticity (STDP) and spiking neural networks.

Research Assistant – VLSI

Indian Institute of Technology (IIT) Guwahati, India

2023 – 2024

INDUSTRIAL AND COLLABORATIVE PROJECTS

Advanced Characterization and Reliability

Taiwan Semiconductor Research Institute (TSRI) project | 2025 – Present

- Conduct endurance, retention, PUND, and charge-pumping measurements; develop pulse-measurement workflows and troubleshoot device behavior with fabrication, process, and equipment teams.

2TnC DRAM Cell Development

Powerchip Semiconductor (PSMC) collaboration | 2025

- Contribute to cell design and process-flow definition, coordinating mask/layout alignment and deposition-related work with integration, lithography, and plasma-etch teams.

FeRAM Chip Tapeout – Ongoing

Industrial Technology Research Institute (ITRI) project | 2025 – Present

- Collaborate on ferroelectric random-access memory (FeRAM) layout, integration, and tapeout preparation using TSRI

EDA 2.0 and Cadence.

SELECTED PUBLICATIONS (5 OF 14)

Publication record: 14 publications comprising 10 journal articles and 4 conference contributions. Selected first- and joint-first-author journal articles are listed below.

1. **“Sub-2 nm Equivalent-Oxide-Thickness Ferroelectric Transistors for Cryogenic Memory and Computing.”**
ACS Nano, 2026. **Joint first author.**
2. **“Endurance Paradox in Hafnium-Oxide-Based Silicon-Channel Ferroelectric Transistors.”**
ACS Applied Materials & Interfaces, 2026. **First author.**
3. **“Vacancy-Assisted Polarization-Catastrophe Framework for Wake-Up and Ferroelectricity in Hafnium Oxide.”**
Nano Letters, 2026. **Joint first author.**
4. **“Long-Term Reliability of Naturally Aged Hafnium-Oxide Ferroelectric Transistors for Energy-Efficient Embedded Memory.”**
Cell Reports Physical Science, 2026. **Joint first author.**
5. **“Spike-Timing-Dependent Learning Dynamics in Silicon-Doped Hafnium-Oxide-Based Ferroelectric Field-Effect Transistors.”**
IEEE Journal of the Electron Devices Society, 2025. **Joint first author.**

TECHNICAL SKILLS

Devices and Reliability: FeFETs, FeRAM, DRAM, hafnium-oxide ferroelectrics, high-k metal-gate (HKMG) stacks, endurance, retention, charge trapping, interface defects, and root-cause analysis.

Electrical Characterization: I-V, C-V, PUND, polarization-voltage (P-V), charge pumping, pulse engineering, high-temperature testing, and cryogenic measurements to 10 K.

Instrumentation: Keysight B1500A with waveform generator/fast measurement unit (WGFMU), Keithley 4200A, Radiant ferroelectric tester, and cryogenic probe station.

Fabrication and Integration: Process-flow definition, thin-film deposition, lithography and plasma/dry-etch process tracking, mask/layout alignment, and memory-device integration.

EDA and Programming: Cadence Virtuoso, KLayout, TSRI EDA 2.0, Python, Verilog, VHDL, Matplotlib, QtiPlot, and Linux.

Cleanroom Qualification: TSRI SM01 and SM01-1 fabrication-facility operation qualifications.

Languages: English (fluent), Bengali (native), Mandarin (basic).

PATENT ACTIVITY

- **Taiwan patent application filed, March 2025:** “Enhanced Independent Dual-Gate Ferroelectric Transistor Memory Using a Decoupled Ferroelectric Capacitor Structure.”
- **Patent filing in preparation, 2026:** “2TnC Ferroelectric Memory Architecture Optimization,” in collaboration with PSMC.

TEACHING AND RESEARCH COMMUNICATION

- Served as a teaching assistant for two graduate/undergraduate courses.
- Presented research at the International Electron Devices and Materials Symposium (IEDMS) and Semiconductor Interface Specialists Conference (SISC).

SCHOLARSHIP MOTIVATION AND CAREER GOALS

My research has strengthened my interest in understanding why memory devices degrade and how interface engineering can improve their reliability. I am applying for the Micron Scholarship to support my doctoral studies and connect my research interests with industrial memory challenges. Having studied and conducted research in India and Taiwan, I value international collaboration and hope to contribute to an inclusive team through knowledge sharing, respectful communication, and cross-disciplinary problem-solving.